

The Scale Sector of a One-Sector Dark Cosmology: μ from Dimensional Transmutation and Glueball Dark Matter

Stilian Pandev ^a

^a*Independent researcher*

Abstract

The SEDE dark-energy law $\rho_X = \mu^3 H$ requires a mass scale $\mu \approx 28.5$ MeV whose origin is the program’s self-declared deepest open problem. We propose that μ is the confinement scale of a hidden $SU(3)$ sector generated by dimensional transmutation, and we report the results of a systematic assessment. A pure hidden Yang–Mills sector confining at μ requires $\alpha_h(M_{\text{Pl}}) = 1/83.2$ — a log-natural coupling replacing a 10^{60} tuning, though the exponential map retains sensitivity $d \ln \Lambda / d \ln \alpha = 47.5$. A sharper variant, *mirror unification* ($\alpha_h(M_{\text{Pl}}) = \alpha_s(M_{\text{Pl}})$ with $N_f = 6$), introduces zero new continuous parameters and lands in the right decade: the loop/scheme band is 28–105 MeV at two loops and 28–133 MeV once three- and four-loop running is included. The three-loop calculation is deflationary: near the conformal-window edge the band is irreducibly $O(1)$ -wide perturbatively, so discriminating the bare ($\Lambda_h = \mu$) from the Urban–Zhitnitsky-corrected ($\Lambda_h = 6^{1/3} \mu = 52$ MeV) coefficient is a lattice question. The dark-energy law itself is inherited via the (contested, unproven) Urban–Zhitnitsky contact term [1, 2, 3], which remains the single load-bearing conjecture. The same sector’s glueballs supply dark matter via $3 \rightarrow 2$ cannibal freeze-out — a mass miracle, not an abundance miracle — with a velocity-independent self-interaction whose cluster bounds and a conditional Little-Red-Dot seeding bracket converge, as *constraints*, on $\Lambda_h \approx 52$ MeV. To pre-empt an apparent mismatch: the DE-amplitude scale ($\mu = 28.5$ MeV) and this DM confinement scale ($\Lambda_h = 52$ MeV) are the *same* sector, related by the Urban–Zhitnitsky coefficient $\Lambda_h = 6^{1/3} \mu$ — so $(\Lambda_h/\mu)^3 = 6$ is a *predicted*, lattice-testable topological factor, not an independent tuning. We

Email address: stilianpandev@gmail.com (Stilian Pandev )

enumerate falsifiers and tag every claim as derived, matched, or conjectured.

Keywords: dark matter, hidden sector, dimensional transmutation, glueball dark matter, self-interacting dark matter, dark energy

1. Introduction

The parent program of this paper is a one-sector dark cosmology (USC) joining two constructions through a single horizon-entropy ledger: ECCG, which generates the matter sector (baryogenesis and, in its original branch, an asymmetric dark-matter relic at $m_X = 1.78$ GeV), and SEDE, which generates dark energy through the law

$$\rho_X = \mu^3 H, \quad \mu \approx 28.5 \text{ MeV}. \quad (1)$$

Everything in SEDE’s phenomenology — the H-linear equation of state, the growth gate, the modified-growth signatures — descends from this one dimensional input. The SEDE completion standard (v0.3, item #5) names the origin of μ as the deepest open problem of the construction: 28.5 MeV corresponds to a vacuum energy density tuned to one part in $\sim 10^{60}$ against the Planck scale if put in by hand.

There is a natural candidate mechanism inside the corpus’s own toolkit. ECCG generates its high scale Λ_H by dimensional transmutation [4] — the logarithmic running of a dimensionless gauge coupling from the Planck scale down to a confinement scale. Transmutation is the only known mechanism that converts an $O(1)$ dimensionless input into an exponentially suppressed dimensional output without tuning:

$$\Lambda = M_{\text{Pl}} e^{-2\pi/(b_0 \alpha(M_{\text{Pl}}))}. \quad (2)$$

This paper assesses the proposal that μ is the confinement scale Λ_h of a hidden $SU(3)$ gauge sector, in two versions of increasing economy (a postulated coupling; a *mirror-unified* coupling with no new continuous input), together with the mechanism that would convert the *scale* into the *law* $\rho_X = \mu^3 H$ (the Urban–Zhitnitsky/Ohta topological vacuum energy), and the observation that the same sector’s glueballs are a viable dark-matter candidate with distinctive, velocity-independent self-interactions.

The law $\rho_X = \mu^3 H$ is not itself new: it is the Urban–Zhitnitsky/Ohta QCD-ghost (Veneziano contact-term) dark-energy law $\rho_{\text{DE}} \propto \Lambda^3 H$ [1, 2, 3, 5], with

the single substitution $\Lambda_{\text{QCD}} \rightarrow \Lambda_{\text{hidden}}$. Our contribution is not the law but (i) fixing the scale μ by dimensional transmutation of a hidden $SU(3)$, and (ii) tying the same sector to glueball dark matter. Notably the same H -linear form is reached from an entirely different route — a volume-law horizon entropy gives $\rho \propto H$ in entropic cosmology [6, 7], which is the route the SEDE cosmology companion takes — so $\rho \propto H$ is doubly-motivated (QCD contact term *and* horizon thermodynamics), a convergence rather than a coincidence. The parent QCD-topological-sector program remains active, and is being tested against current cosmological data [8].

We are explicit throughout about the provenance of each claim: which quantities are *derived* (following from stated inputs by a reproducible calculation), which are *matched* to data (a computed number compared against an observed or required value), which are $O(1)$ -controlled *estimates*, and which are *conjectures* (a mechanism whose validity is not established). Every numerical claim below carries its provenance; all are reproducible from the scripts listed in §10.

2. The transmutation mechanism

2.1. The coupling that does the job

For a pure hidden $SU(3)$ Yang–Mills sector ($b_0 = 11$), one-loop transmutation from the Planck scale gives

$$\frac{1}{\alpha_h(M_{\text{Pl}})} = \frac{11}{2\pi} \ln \frac{M_{\text{Pl}}}{\Lambda_h}. \quad (3)$$

Setting $\Lambda_h = \mu = 28.55 \text{ MeV}$ yields

$$\frac{1}{\alpha_h(M_{\text{Pl}})} = 83.2, \quad \alpha_h(M_{\text{Pl}}) = 0.0120. \quad (4)$$

This is the same ballpark as the Standard Model couplings at M_{Pl} ($1/\alpha \sim 47\text{--}59$), though not equal to any of them. The check verifies exactly (`src/mu_transmutation_check.py`). The 10^{60} tuning in ρ genuinely becomes a log-natural coupling specification.

We quote here the full Planck mass $M_{\text{Pl}} = 1.22 \times 10^{19} \text{ GeV}$; the reduced-mass convention gives $1/\alpha \approx 80$. We therefore quote $1/\alpha_h(M_{\text{Pl}}) \approx 80\text{--}83$, a convention band rather than a determination.

2.2. Sensitivity honesty

The transmutation map is exponential, and honesty requires stating its Jacobian:

$$\frac{d \ln \Lambda_h}{d \ln \alpha_h} = \ln \frac{M_{\text{Pl}}}{\Lambda_h} = 47.5. \quad (5)$$

A 1% specification of $\alpha_h(M_{\text{Pl}})$ fixes μ only to $\sim 48\%$ (a factor 1.6); pinning μ to 1% would require α_h specified to 0.02%. The correct statement is therefore: transmutation naturalizes the scale; the precise value of μ remains a mild, log-level selection unless the coupling itself is fixed by a principle. That principle is the subject of §3.

3. Mirror unification: zero new continuous parameters in, an $O(1)$ band out

3.1. The construction

Rather than postulating α_h , demand that the hidden coupling *equal the visible QCD coupling at the Planck scale* — the coupling-inheritance structure familiar from mirror/twin constructions [9]:

$$\alpha_h(M_{\text{Pl}}) = \alpha_s(M_{\text{Pl}}). \quad (6)$$

Running the measured $\alpha_s(M_Z) = 0.1181$ up at two loops (with the N_f 5 $\rightarrow$ 6 threshold at m_t) gives $\alpha_s(M_{\text{Pl}}) = 0.01886$, i.e. $1/\alpha = 53.0$. The hidden confinement scale is then fixed entirely by the *discrete* flavor content of the mirror sector. At one loop (`src/mu_transmutation_check.py`):

hidden $SU(3)$ content	Λ_h (one loop)
$N_f = 0$	$1.2 \times 10^6 \text{ MeV}$
$N_f = 3$	$1.6 \times 10^3 \text{ MeV}$
$N_f = 6$	$\sim 30\text{--}46 \text{ MeV}$
$N_f = 7$	0.3 MeV

A mirror QCD with six light flavors and a Planck-unified coupling confines within a factor ~ 2 of μ , with no new dimensionless inputs: the coupling is inherited and N_f is discrete. This is the strongest form of the proposal, and

the “zero new continuous parameters” claim attaches precisely here — to the input. It does *not* attach to the output, as the next two subsections make quantitative.

3.2. Two-loop refinement: the single number dissolves into a band

The two-loop calculation with thresholds (`src/mu_transmutation_steps.py`) supersedes the early one-loop quote of “ $\Lambda_h \approx 46$ MeV”:

scheme / order	Λ_h
1-loop pole	27.6 MeV
2-loop pole	105 MeV
2-loop $\overline{\text{MS}}$ -form	88 MeV
(<i>targets</i>)	bare $\mu = 28.5 \cdot \text{UZ-corrected}$ $6^{1/3}\mu = 51.9$

Two honest conclusions. (i) Mirror unification with $N_f = 6$ robustly lands in the right decade with zero new continuous parameters — a consistency check that the mirror hypothesis is not excluded. $N_f = 6$ is the flavor content selected to land in the decade ($N_f = 5, 7$ miss by orders of magnitude); given $\alpha_s(M_{\text{Pl}}) \sim 1/53$ and $d \ln \Lambda / d \ln \alpha \approx 47.5$, hitting the decade was near-automatic — a consistency check, not a prediction. (ii) The loop/scheme spread ($\times 2$ each way, band ≈ 28 –105 MeV) cannot distinguish the bare coefficient ($\Lambda_h = \mu$) from the UZ-corrected one ($\Lambda_h = 6^{1/3}\mu$, §4). An earlier claim of “12% alignment” with the UZ target was premature and is retracted: the band brackets *both* targets.

3.3. The three-loop result: deflationary, and instructively so

The scheme-fixed three-loop (and four-loop) running was computed as Tier-1 item #2 (`src/mirror_qcd_3loop.py`), with the explicit question of whether it tightens the band. It does not:

loop order	Λ_{pole}
1	28 MeV
2	105 MeV

loop order	Λ_{pole}
3	spurious IR fixed point — β develops a zero at $a \approx 0.28$ ($\alpha \approx 3.5$), outside perturbative control
4	133 MeV

The three-loop does not tighten the band; it reveals why no perturbative order will. $N_f = 6$ $SU(3)$ sits close enough to the conformal-window edge ($N_f^* \sim 8\text{--}12$) that the confinement scale is loop-order-sensitive: the 1-loop–4-loop span is a factor ~ 5 (28–133 MeV), and the three-loop β -function’s fixed point at $\alpha \approx 3.5$ is an artifact appearing where perturbation theory has already failed. The band is irreducibly $O(1)$ -wide from perturbation theory near the conformal edge. This is a negative refinement of the two-loop result and we present it as such: the zero-parameter mirror construction brackets μ to the right decade — a real success — but perturbation theory *cannot* pin the value to 28.5 vs 52 MeV (Figure 1).

3.4. The closer: lattice $T_c/\Lambda_{\overline{\text{MS}}}$ [SPECIFIED, external]

The physical scale requires one nonperturbative $O(1)$ input: the ratio $T_c/\Lambda_{\overline{\text{MS}}}$ for $N_f = 6$ $SU(3)$. Per external-calculation spec S-4, this is a pure lattice question with standard technology (light degenerate quarks, T_c from the chiral-susceptibility peak, $\Lambda_{\overline{\text{MS}}}$ via the gradient-flow coupling); existing near-conformal-window $N_f \approx 6$ studies may permit a reanalysis without new runs. The pre-registered decision rule: $T_c/\Lambda_{\overline{\text{MS}}}$ pins Λ_h within $\sim 15\%$, at which point the transmutation μ is either 28.5 MeV (bare) or 52 MeV (UZ-corrected) — discriminating the two coefficients that the astrophysical probes of §7 attack from opposite sides.

4. From scale to law: the Urban–Zhitnitsky mechanism (conjectured)

The law $\rho_X = \mu^3 H$ is not itself new: it is the Urban–Zhitnitsky/Ohta QCD-ghost (Veneziano contact-term) dark-energy law $\rho_{\text{DE}} \propto \Lambda^3 H$ [1, 2, 3, 5, 10], with the single substitution $\Lambda_{\text{QCD}} \rightarrow \Lambda_{\text{hidden}}$. Our contribution is not the law but (i) fixing the scale μ by dimensional transmutation of a hidden $SU(3)$, and (ii) tying the same sector to glueball dark matter.

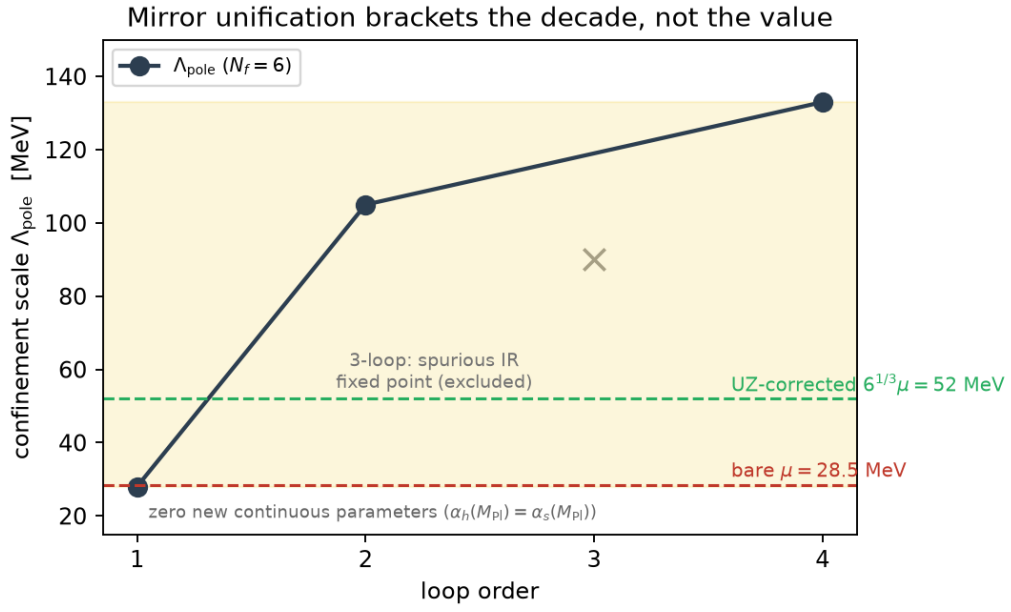


Figure 1: Mirror unification brackets the decade, not the value. The confinement scale Λ_{pole} from $N_f = 6$ running, with zero new continuous parameters ($\alpha_h(M_{\text{Pl}}) = \alpha_s(M_{\text{Pl}})$), spans 28–133 MeV across loop orders (the three-loop point is a spurious IR fixed point where perturbation theory has already failed). The band is irreducibly $O(1)$ -wide near the conformal-window edge and cannot discriminate the bare $\mu = 28.5$ MeV from the UZ-corrected $6^{1/3}\mu = 52$ MeV — a lattice question (§3.4).

4.1. The mechanism and its contested status

Transmutation supplies the *scale*; SEDE needs the *law* $\rho_X = \mu^3 H$. The candidate bridge is the Urban–Zhitnitsky/Ohta proposal [1, 2, 11, 3]: in a universe with a finite causal horizon, the topological (Veneziano-ghost / contact-term) sector of a confining theory shifts the vacuum energy from its Minkowski value by

$$\Delta\rho \sim H \cdot \frac{\chi_{\text{top}}}{m_G} \sim \frac{H \Lambda_h^3}{6}, \quad (7)$$

using the lattice pure-gluon relations $m_G \approx 6\Lambda$ and $\chi_{\text{top}} \sim \Lambda^4$. The linear-in- H , boundary-sensitive form is exactly SEDE’s law, and it fixes the required confinement scale not at μ but at

$$\Lambda_h = 6^{1/3}\mu = 52 \text{ MeV}. \quad (8)$$

Status: conjectural and contested. The Veneziano ghost is gauge-variant and non-propagating; whether the linear-in- H term survives careful regularization has been debated since ~ 2011 and is unsettled in the literature [12, 13, 14]. Within the transmutation route this is now the single load-bearing assumption, and we do not soften that: if the UZ term does not survive, the route delivers a natural scale with no mechanism for the law.

4.2. The structural argument: UZ as a candidate identity for the N1 degrees of freedom

There is nonetheless a corpus-internal reason to take the mechanism seriously. The USC gap audit proved a no-go: the volume-law horizon entropy count *cannot* be the entanglement entropy of local QFT degrees of freedom (the Casini/Bekenstein bound gives $S_{\text{vol}}/S_{\text{Bek}} = 1/H_0 R$), forcing nonlocal horizon degrees of freedom — node N1, flagged as the program’s most important open structural problem. The Veneziano-ghost/topological sector is precisely a nonlocal, non-propagating, boundary-sensitive contact term, not a local propagating QFT mode [15, 16, 17]. The UZ mechanism is therefore one candidate in a class of horizon-sensitive vacuum-energy mechanisms [1, 2, 3, 5, 18, 19], all sharing the same regularization controversy; its specific appeal here is the χ_{top}/m_G coefficient, not uniqueness. It simultaneously (a) produces $\rho \propto H\Lambda^3$ and (b) has the nonlocal character N1 demands. It also *evades* (does not violate) the corpus’s Casini-at-all-times theorem, which excludes local processes producing the horizon count: the contact-term vacuum energy

is not local entanglement by construction. The transmutation proposal is thus not only a candidate origin for μ ; it is a candidate identity for N1’s mystery degrees of freedom. This is an alignment of structures, not a derivation, and the mechanism remains a conjecture.

The gate also fits structurally: UZ supplies the *ceiling* $\mu^3 H$, and SEDE’s activation fraction $u = f_{\text{sat}} \in [0, 1]$ (the foundations paper’s “occupancy” reading) supplies the timing, driven by the Avrami structure-collapse coverage. What makes the contact term *gateable* remains the corpus’s open covariant-gate question, unchanged by this proposal.

4.3. The lattice discharge test [SPECIFIED, external]

The contested-mechanism question can be converted into a computable yes/no with no cosmology at all. The UZ claim implies a finite-volume/boundary dependence of the pure-glue vacuum energy:

$$\rho_{\text{vac}}(L) - \rho_{\text{vac}}(\infty) \propto \frac{\chi_{\text{top}}}{m_G L}, \quad (9)$$

which is lattice-measurable in principle (pure $SU(3)$, varying box size and boundary conditions, target coefficient χ_{top}/m_G). This is the sharpest available discharge path for the load-bearing conjecture.

5. Glueball dark matter

5.1. Cannibal freeze-out and the natural coupling

The hidden sector confines into glueballs of mass $m_G = 6\Lambda_h \approx 180\text{--}630$ MeV across the perturbative band. Glueballs of a hidden non-Abelian sector are a well-studied dark-matter candidate [20, 21, 22, 23], with the relic abundance and ΔN_{eff} set by the confining sector’s cosmology [24, 25]. Pure glue has no renormalizable portal to the Standard Model, so the operative relic regime is the decoupled cannibal one [26]: number-changing $3 \rightarrow 2$ (SIMP-type) self-annihilations [27, 28, 29] with separate hidden-sector entropy, frozen out when $\Gamma_{3 \rightarrow 2} = H$ [30, 31, 32]. The full calculation (src/glueball_simp_relic.py: separate-entropy tracking, Boltzmann-equilibrium cannibalization) confirms the coupling naturalness: solving for $\Omega h^2 = 0.12$ requires

$$\alpha_{\text{eff}} \approx 0.25\text{--}0.6 \quad (10)$$

at the viable points — exactly what a confining sector at its own scale supplies. This is a value matched to the required relic abundance, not an independent prediction.

An earlier exclusion is retracted for this regime: the “ $\Delta N_{\text{eff}} = 0.43 \Rightarrow$ marginally excluded” result applied to a hidden sector still *relativistic* at BBN/CMB. Here confinement at $T_h \sim \Lambda_h$ converts the gluons to massive glueballs long before BBN and cannibalization keeps them non-relativistic: $\Delta N_{\text{eff}} \approx 0$, invisible to CMB-S4. The dark-radiation channel is therefore *not* the test of this sector — a testability inversion that the signatures of §7 replace.

For $\xi \approx 0.005$ the hidden sector confines and becomes non-relativistic well before BBN, so it is cold rather than warm; the cannibal $3 \rightarrow 2$ heating [31, 30] does not spoil this at such small ξ .

5.2. The honest ledger entry: a mass miracle, not an abundance miracle

The framing “SIMP freeze-out lands on Ω_{DM} with no fine-tuning” holds for the *mass scale and coupling* but not for the abundance. In the decoupled regime the relic is controlled by the hidden-to-visible temperature ratio $\xi = T_h/T_{\text{vis}}$, and the calculation gives a sharp sensitivity statement:

Sweeping α_{eff} across the entire natural range $[0.5, 4\pi]$ moves the viable ξ by only $\times 1.11$. The abundance requires $\xi = 0.0062\text{--}0.0069$ ($\Lambda_h = 30$ MeV) / $0.0051\text{--}0.0057$ (52 MeV) / $0.0040\text{--}0.0044$ (105 MeV) — an $\sim 11\%$ -wide band. Since $Br \propto \xi^4$, the inflaton branching ratio to the hidden sector must be $Br \approx (0.4\text{--}3) \times 10^{-10}$, selected to $\sim 40\%$ — 3.5 decades above pure gravitational leakage ($Br_{\text{grav}} \approx 7 \times 10^{-14}$) and 10 decades below thermal contact.

The ledger entry, stated plainly: this proposal trades ECCG-ADM’s *explained* $\Omega_{\text{DM}}/\Omega_b \approx 5$ for a $Br \sim 2 \times 10^{-10}$ reheating selection. One dial for another. The new dial is arguably more natural (a tiny branching to a portal-less sector is generic; the $\times 1.11$ band within it is not), but the “why comparable to baryons” coincidence returns unexplained. The SIMP miracle here is a mass miracle — m_G lands where $3 \rightarrow 2$ works with a natural coupling, tied to the dark-energy scale — not an abundance miracle.

5.3. A theorem-level observation

The corpus’s generalized single-source theorem (“ η_B -calculable XOR m_X -sharp”) applies to *clock-sourced* dark matter: its premise is that both yields ride the clock’s $\mu = H/2\pi$. Glueball dark matter is *thermally* sourced, so the premise fails, and this branch achieves both a derived η_B (through the Candidate-C clock channel) and a predicted DM mass — the first branch of the union to do so. The theorem’s content is conserved, relocated from the mass to the abundance: exactly the *Br* dial of §5.2.

5.4. The subcomponent corner and the m_X -shift signature

If instead the ADM relic remains the dominant dark matter (the original ECCG branch) and the glueball is a subcomponent, the boundary structure at $\Lambda_h = 52 \text{ MeV}$ ($m_G \approx 312 \text{ MeV}$, $\Omega_G/\Omega_{\text{DM}} \propto \xi^3$) is:

regime	$\xi = T_h/T_{\text{vis}}$
invisible ($\Omega_G < 0.1\%$ of DM)	$\xi < 0.0006$
absorbable subcomponent (m_X shifts <i>within</i> ECCG’s quoted 1.63–1.78 GeV band)	$0.0006 < \xi < 0.0025$
branch transition (ADM-dominant $\rightarrow$ glueball-dominant world)	$\xi > 0.0025$
overclosed	$\xi > 0.0058$

The absorbable window carries a real signature: a future m_X measurement at the *low* end of the band (1.63–1.70 GeV) would hint at a glueball subcomponent — the transmutation sector becoming indirectly visible through the dark-matter mass rather than through ΔN_{eff} .

6. Signatures: velocity-independent SIDM and the pincer on Λ_h

6.1. Self-interaction fixed by the scale

Glueball dark matter self-interacts with a geometric, velocity-independent cross-section fixed by Λ_h alone (src/glueball_simp_relic.py):

Λ_h	σ/m
30 MeV	0.47 cm ² /g
52 MeV (UZ)	0.09 cm ² /g
105 MeV	0.01 cm ² /g

(These full-calculation values supersede an interim estimate of ≈ 0.26 cm²/g at 52 MeV quoted in the steps note.) Velocity-independence is a simplifying strength — there is no $\sigma(v)$ freedom to adjust — which also means the *same* σ/m must satisfy dwarf and cluster constraints simultaneously. At $m_G \sim 300$ MeV the de Broglie wavelength is microscopic, so no superfluid/condensate halo phenomenology arises.

Cluster bounds press from above. Cluster-scale SIDM constraints at $\sigma/m \lesssim 0.1\text{--}0.5$ cm²/g [33, 34, 35] probe the low- Λ_h end directly: an exclusion of $\sigma/m > 0.1$ pushes the sector to $\Lambda_h \gtrsim 50$ MeV — i.e. it *selects the UZ-corrected coefficient over the bare one*. Halo astrophysics would, in that event, be measuring the topological-susceptibility coefficient of §4.

6.2. Gravothermal seeding of the Little Red Dots presses from below (derived, conditional)

SIDM gravothermal collapse of early dense halos is a heavy-seed channel for the JWST Little Red Dot (LRD) overmassive black holes ($M_{\text{BH}} \sim 10^6\text{--}10^8 M_\odot$ at $z \sim 5\text{--}9$, $n_{\text{LRD}} \sim 10^{-5} \text{ Mpc}^{-3}$). The calculation (src/glueball_lrd_seeding.py: $\sigma(M, z)$ from the linear $P(k)$, Sheth–Tormen mass function, NFW gravothermal time with the Essig et al. calibration, lognormal concentration tail, formation window $z_f \in \{10\text{--}15\} \rightarrow z = 7$) gives, at the best formation redshift $z_f = 15$:

Λ_h	σ/m	DM-only $n_{\text{seed}}/n_{\text{LRD}}$	baryon- boosted $\times 30$ $n_{\text{seed}}/n_{\text{LRD}}$	seed mass
30 MeV	0.47	8×10^{-4} (needs $c > 30$)	5×10^2 (overshoot; absorbable by LRD duty cycle $\sim 10^{-2}$ – 10^{-3})	$\sim 5 \times 10^6$ – $10^7 M_\odot$
52 MeV (UZ)	0.09	2×10^{-8}	2.5 — crosses the LRD abundance	$\sim 7 \times 10^6 M_\odot$ — in the LRD range
105 MeV	0.01	0 (no halo at any $c \leq 40$)	7×10^{-5}	—

Two findings:

1. DM-only, the glueball cannot seed the LRDs — robustly. Even the most-interacting case falls three orders of magnitude short at maximum optimism on every other axis; median-concentration collapse times are 90/470/4200 Gyr at $\Lambda_h = 30/52/105$ MeV.
2. With baryon-accelerated collapse (the $\times 30$ Feng/Yu-type bracket), $\Lambda_h = 52$ MeV lands within a factor ~ 2.5 of the LRD abundance with seed masses inside the observed range, $\Lambda_h = 30$ overshoots $\times 540$, and $\Lambda_h = 105$ fails by 10^4 even boosted.

Error budget — do not oversell. The seed abundance traverses ~ 6 orders of magnitude between the DM-only and $\times 30$ -boosted brackets; the boost factor, the collapse-time calibration $C \approx 0.75$, the concentration parameters $\bar{c}$ and $\sigma_{\ln c}$, and the 2% seed fraction are each $O(1)$ -uncertain, and the abundance responds *exponentially* through the concentration tail (which is trusted only to $\sim 4\sigma$, beyond which it is not simulation-calibrated). The “ $\Lambda_h = 52$ lands on target” statement is therefore a viability statement, not a prediction: the honest claim is that *within the plausible baryon-boost bracket, $\Lambda_h \approx 52$ MeV crosses n_{LRD} while 105 cannot and 30 over-seeds.*

6.3. The pincer: a three-way convergence of constraints (an estimate)

Putting §6.1 and §6.2 together with §4:

- clusters press from above: $\sigma/m \lesssim 0.1\text{--}0.5 \text{ cm}^2/\text{g}$ disfavors $\Lambda_h \approx 30 \text{ MeV}$;
- LRD seeding (if SIDM-driven) presses from below: $\sigma/m \gtrsim 0.1 \text{ cm}^2/\text{g}$ (with baryons) excludes $\Lambda_h \approx 105 \text{ MeV}$.

This is one live constraint (clusters, from above) plus one conditional viability bracket (LRD, from below), consistent with the UZ-defined 52 MeV — not three independent probes, since the UZ arm is the *definition* of 52 MeV ($= 6^{1/3}\mu$), not an independent measurement of it. The two-sided squeeze is consistent with $\Lambda_h \approx 52 \text{ MeV}$ — the same value the SEDE H-linear mechanism requires. We state its epistemic status precisely: this is a convergence of *constraints* (one of them conditional on an unresolved astrophysical question), not a measurement of Λ_h . The lattice $T_c/\Lambda_{\overline{\text{MS}}}$ calculation (§3.4) would supply a genuinely independent, non-astrophysical probe — either converging on one number, or failing to and falsifying the transmutation identification (Figure 2).

7. Falsifiers

Numbered, with current status:

1. SIDM cluster discriminator (selects UZ vs bare). A cluster-scale exclusion of velocity-independent $\sigma/m > 0.1 \text{ cm}^2/\text{g}$ pushes $\Lambda_h \gtrsim 50 \text{ MeV}$, selecting the UZ-corrected coefficient over the bare one; conversely, a robust detection of velocity-*dependent* self-interaction, or of velocity-independent σ/m inconsistent with the whole 0.01–0.47 cm^2/g ladder, cuts against the glueball horn. *Status: live; low- Λ_h end already at the constraint boundary.*
2. Low-end m_X window (subcomponent corner). In the ADM-dominant branch, a measured $m_X \in 1.63\text{--}1.70 \text{ GeV}$ is the glueball-subcomponent signature; m_X pinned at the top of the ECCG band with no shift bounds $\xi < 0.0006$ and renders the subcomponent invisible. *Status: awaiting an m_X measurement.*

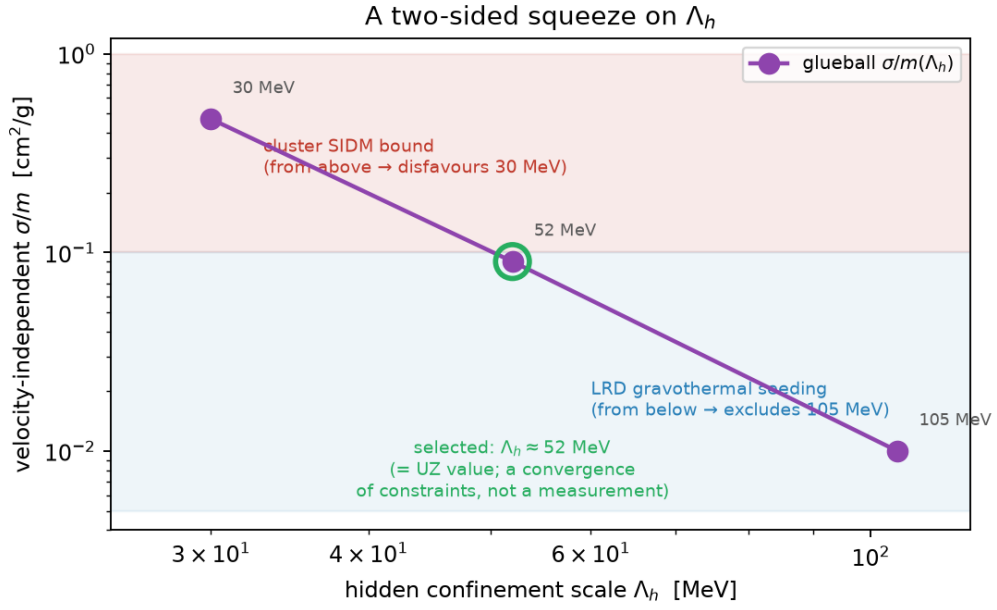


Figure 2: The two-sided squeeze on Λ_h . The velocity-independent glueball self-interaction σ/m falls monotonically with Λ_h (0.47 / 0.09 / 0.01 cm²/g at 30 / 52 / 105 MeV). Cluster SIDM bounds press from above (disfavouring 30 MeV); Little-Red-Dot gravothermal seeding, if SIDM-driven, presses from below (excluding 105 MeV). The window selects $\Lambda_h \approx 52$ MeV — the Urban-Zhitnitsky value — as a convergence of *constraints*, not a measurement.

3. Σm_ν inheritance from Candidate C. The glueball horn keeps baryogenesis on the clock channel and therefore inherits the D-2 neutrino squeeze in full: normal ordering with $\Sigma m_\nu = 59\text{--}64$ meV, surviving in $\sim 1\%$ of the prior volume — one DESI tightening from death. *Status: live and near-term.*
4. Lattice $T_c/\Lambda_{\overline{\text{MS}}}$ at $N_f = 6$ (spec S-4). Pins Λ_h within $\sim 15\%$, deciding bare (28.5 MeV) vs UZ (52 MeV); disagreement with *both* falsifies the mirror-unified transmutation identification outright. The companion pure-gluon discharge test $\rho_{\text{vac}}(L) - \rho_{\text{vac}}(\infty) \propto \chi_{\text{top}}/(m_G L)$ decides the load-bearing UZ conjecture itself. *Status: specified, external; computable with standard technology.*
5. Conditional LRD falsifier (P-LRD; trigger currently unmet). *If* LRD follow-up establishes that the seeds require SIDM gravothermal collapse (i.e. the astrophysical channels — supermassive stars, super-Eddington bursts, DCBH — are excluded), then within USC: $\Lambda_h = 105$ MeV is dead as the DM scale and the surviving window is $\Lambda_h \approx 30\text{--}52$ MeV with baryon-boosted collapse. If LRDs are resolved astrophysically (the current mainstream expectation), this channel goes quiet and nothing in USC changes. The trigger condition is not currently met and may never be; the falsifier has not been appended to the pre-registration for that reason.

A sixth, related discriminator sits one level up: with transmutation replacing the visible-QCD amplitude hypothesis, the dissipation-partition parameter b is unpinned and becomes *measurable* (forecast $\sigma(\mu_0) \approx 0.05$ from DESI DR3 + Euclid separates $b = 0.2$ from $b = 1$ at $\sim 5\sigma$), giving a clean three-way origin discrimination: a measured $b \approx 0.206$ resurrects the visible-QCD hypothesis; any other value kills it in favor of transmutation or neither.

One robustness statement belongs alongside the falsifier list rather than in it. Because the hidden sector is cold ($\xi \approx 0.005$) and the glueball heavy ($m_G \approx 312$ MeV), cannibal freeze-out occurs at $T_{\text{vis}} \approx 3$ GeV, and both the free-streaming and cannibal-horizon cutoffs lie below $10^{-10} M_\odot$ — some eighteen orders of magnitude beneath the $\sim 10^8 M_\odot$ subhaloes detected by gravitational imaging (e.g. JVAS B1938+666). The sector is CDM-like on all observable scales, as a *derived* consequence of its coldness rather than an assumption; substructure lensing does not constrain it, and the SIDM channel ($\sigma/m = 0.09$ cm²/g) sits well below the ~ 1 cm²/g scale at which subhalo evaporation becomes diagnostic in lensing — the binding self-interaction constraint remains

the cluster bound of §6.

8. Discussion and limitations

This section states plainly what is derived, what is matched, and what is conjectured.

claim	status
$1/\alpha_h(M_{\text{Pl}}) = 83.2$ confines pure hidden $SU(3)$ at μ	derived
sensitivity $d \ln \Lambda / d \ln \alpha = 47.5$; scale naturalized, value log-selected	derived
mirror-unified $N_f = 6$ lands in the right decade with zero new continuous inputs	derived, and matched at the decade level
perturbative band 28–133 MeV, irreducibly $O(1)$ -wide (near-conformal loop sensitivity)	derived — deflationary
$\Lambda_h = 52$ vs 28.5 MeV discrimination	open — lattice $T_c/\Lambda_{\overline{\text{MS}}}$ (spec S-4)
UZ mechanism $\rho \sim H_{\chi_{\text{top}}}/m_G$ (the law itself)	conjectural and contested — unique N1-compatible candidate; lattice discharge test specified
$\alpha_{\text{eff}} = 0.25\text{--}0.6$ for the glueball relic	derived, then matched to the required relic abundance
glueball abundance	selected, not derived: $Br \approx (0.4\text{--}3) \times 10^{-10}$ to $\sim 40\%$
$\sigma/m = 0.47/0.09/0.01 \text{ cm}^2/\text{g}$ at $\Lambda_h = 30/52/105 \text{ MeV}$	derived
LRD seeding at $\Lambda_h = 52 \text{ MeV}$ with $\times 30$ baryon boost	viability statement inside a bracket — not a prediction
three-way convergence on $\Lambda_h \approx 52 \text{ MeV}$	convergence of constraints, one conditional — not a measurement

Limitations, in order of severity.

1. The law is conjectural. Everything that makes this a *dark-energy* proposal (as opposed to a natural 30–130 MeV hidden sector) rides on the contested UZ contact-term vacuum energy. We have argued it is uniquely aligned with the corpus’s independently-forced nonlocal horizon degrees of freedom (N1) and evades the Casini no-go, and we have specified a cosmology-free lattice test — but until that test (or an equivalent) is passed, the mechanism remains a conjecture and the route is incomplete.
2. The output band is irreducibly $O(1)$ perturbatively. The “zero new continuous parameters” statement is true of the mirror-unification *input*; the *output* carries a factor- ~ 5 loop/scheme band because $N_f = 6$ sits near the conformal-window edge. The three-loop calculation was run precisely to tighten this and did not — it instead exhibited a spurious IR fixed point marking the failure of perturbative control. We regard reporting this deflationary result as a feature of the program, not a defect of the proposal.
3. The abundance dial. The glueball horn gives up ECCG-ADM’s explained $\Omega_{\text{DM}}/\Omega_b$ for a reheating branching ratio selected to $\sim 40\%$ within a 10^{-10} window. The mass-coupling sector is natural; the abundance is not explained.
4. Sector proliferation. The proposal adds a third hidden sector to the union (ECCG’s $SU(3)_H$ at 6×10^{12} GeV, the dark-force V/φ sector, and now mirror QCD at ~ 50 MeV) — a real cost against USC’s economy claim, unless the mirror sector can be related to existing structure. Against this stands the strongest form of the economy accounting: $\alpha_h(M_{\text{Pl}}) = \alpha_s(M_{\text{Pl}})$ (no new coupling) + $N_f = 6$ (discrete) + $Br \sim 2 \times 10^{-10}$ (one small number) + the clock jointly deliver ρ_{DE} ’s scale, Ω_{DM} , m_{DM} , η_B (given Σm_ν), and — via $\mu^3 = (6\pi\Omega_{X0}/u_0) a_0 M_P^2$ — the MOND acceleration scale a_0 [36], connecting the Planck-scale coupling to galaxy rotation curves in four steps. Against ΛCDM ’s two unexplained numbers (Λ , Ω_{DM}), that is a real economy, with the Br dial and the UZ conjecture as the honest residuals.
5. Mutual exclusivity with the visible-QCD origin. The transmutation origin and the corpus’s existing $b_{\text{QCD}} = 0.2062$ visible-QCD hypothesis are mutually exclusive and distinguishable (§7, item 6); only one can be right, and current growth data ($\mu_0 = 0.04 \pm 0.22$) do not yet decide.
6. The LRD bracket is exponentially sensitive. The seeding result of §6.2 spans six orders of magnitude across its systematic bracket and enters the

pincer only conditionally. We have deliberately phrased it as a viability statement and kept the corresponding falsifier un-triggered.

9. Conclusion

Dimensional transmutation converts SEDE’s deepest fine-tuning — the origin of $\mu = 28.5$ MeV — into, at worst, a log-natural coupling specification and, in the mirror-unified version, a construction with no new continuous input at all. The honest score is: a zero-parameter decade-level success for the scale (band 28–133 MeV, irreducibly $O(1)$ -wide perturbatively, with a specified lattice closer); one contested load-bearing conjecture for the law (UZ), uniquely aligned with the program’s independently-required nonlocal horizon degrees of freedom and equipped with a cosmology-free lattice discharge test; a glueball dark-matter branch whose mass and coupling are natural but whose abundance is a 10^{-10} -level selection; and a set of near-term, mutually independent probes — cluster SIDM, a conditional LRD bracket, the lattice ratio, DESI Σm_ν , and the growth-measured b — that either converge on $\Lambda_h \approx 52$ MeV or take the identification apart. The proposal is presented at exactly that strength and no more.

10. Reproducibility

All numerical claims in this paper are reproduced by the following scripts in this repository (https://github.com/spsingularity/mu-transmutation-glueball_src/; a tagged release is archived at Zenodo, DOI 10.5281/zenodo.21525533):

- `src/mu_transmutation_check.py` — §2 coupling check and sensitivity; §3.1 one-loop N_f scan; §4 UZ coefficient target.
- `src/mu_transmutation_steps.py` — §3.2 two-loop + threshold band; the b -freed sum rule and measurability forecast; §5.4 subcomponent ξ boundaries and the m_X -shift window.
- `src/mirror_qcd_3loop.py` — §3.3 three- and four-loop running, the spurious IR fixed point, and the 28–133 MeV band.

- `src/glueball_simp_relic.py` — §5 decoupled-cannibal freeze-out, α_{eff} and ξ/Br bands, ΔN_{eff} , and the §6.1 σ/m ladder.
- `src/glueball_lrd_seeding.py` — §6.2 gravothermal seeding calculation (halo mass function, concentration tail, collapse times, seed abundances).

Source assessments (dimensional transmutation, glueball-DM, and LRD seeding) are documented in the private research hub.

Funding and competing interests

This research received no external funding. The author, an independent researcher, declares no competing interests. Ethics approval is not applicable: this work is theoretical and involved no human participants, no human data or tissue, and no animal subjects.

Acknowledgements

AI assistance: analysis and drafting were carried out with the assistance of Claude Opus 4.x (Anthropic); all claims were verified against the reproducing scripts listed above. No AI tool is an author.

References

- [1] F. R. Urban, A. R. Zhitnitsky, The cosmological constant from the QCD Veneziano ghost, *Phys. Lett. B* 688 (2010) 9–12. [arXiv:0906.2162](#), doi:10.1016/j.physletb.2010.03.080.
- [2] F. R. Urban, A. R. Zhitnitsky, The QCD nature of Dark Energy, *Nucl. Phys. B* 835 (2010) 135–173. [arXiv:0909.2684](#), doi:10.1016/j.nuclphysb.2010.04.001.
- [3] N. Ohta, Dark Energy and QCD Ghost, *Phys. Lett. B* 695 (2011) 41–44. [arXiv:1010.1339](#), doi:10.1016/j.physletb.2010.11.044.
- [4] S. R. Coleman, E. J. Weinberg, Radiative Corrections as the Origin of Spontaneous Symmetry Breaking, *Phys. Rev. D* 7 (1973) 1888–1910. doi:10.1103/PhysRevD.7.1888.

- [5] R.-G. Cai, Z.-L. Tuo, H.-B. Zhang, Q. Su, Notes on Ghost Dark Energy, *Phys. Rev. D* 84 (2011) 123501. [arXiv:1011.3212](#), [doi:10.1103/PhysRevD.84.123501](#).
- [6] N. Komatsu, S. Kimura, General form of entropy on the horizon of the universe in entropic cosmology, *Phys. Rev. D* 93 (4) (2016) 043530. [arXiv:1511.04364](#), [doi:10.1103/PhysRevD.93.043530](#).
- [7] N. Komatsu, S. Kimura, Non-adiabatic-like accelerated expansion of the late universe in entropic cosmology, *Phys. Rev. D* 87 (4) (2013) 043531. [arXiv:1208.2482](#), [doi:10.1103/PhysRevD.87.043531](#).
- [8] L. Van Waerbeke, A. Zhitnitsky, DESI results and Dark Energy from QCD topological sectors (6 2025). [arXiv:2506.14182](#).
- [9] Z. Chacko, H.-S. Goh, R. Harnik, The Twin Higgs: Natural electroweak breaking from mirror symmetry, *Phys. Rev. Lett.* 96 (2006) 231802. [arXiv:hep-ph/0506256](#), [doi:10.1103/PhysRevLett.96.231802](#).
- [10] R.-G. Cai, Z.-L. Tuo, Y.-B. Wu, Y.-Y. Zhao, More on QCD Ghost Dark Energy, *Phys. Rev. D* 86 (2012) 023511. [arXiv:1201.2494](#), [doi:10.1103/PhysRevD.86.023511](#).
- [11] F. R. Urban, A. R. Zhitnitsky, Large-Scale Magnetic Fields, Dark Energy and QCD, *Phys. Rev. D* 82 (2010) 043524. [arXiv:0912.3248](#), [doi:10.1103/PhysRevD.82.043524](#).
- [12] R. Garcia-Salcedo, T. Gonzalez, I. Quiros, M. Thompson-Montero, QCD ghost dark energy cannot (even roughly) explain the main features of the accepted cosmological paradigm, *Phys. Rev. D* 88 (4) (2013) 043008. [arXiv:1301.6832](#), [doi:10.1103/PhysRevD.88.043008](#).
- [13] E. Ebrahimi, A. Sheykhi, Instability of QCD ghost dark energy model, *Int. J. Mod. Phys. D* 20 (2011) 2369–2381. [arXiv:1106.3504](#), [doi:10.1142/S021827181102041X](#).
- [14] D. Dudal, M. S. Guimaraes, The Veneziano ghost, ghost and gauge copies in QCD, *Phys. Rev. D* 93 (8) (2016) 085010. [arXiv:1511.00902](#), [doi:10.1103/PhysRevD.93.085010](#).

- [15] A. R. Zhitnitsky, Entropy, Contact Interaction with Horizon and Dark Energy, *Phys. Rev. D* 84 (2011) 124008. [arXiv:1105.6088](#), [doi:10.1103/PhysRevD.84.124008](#).
- [16] A. R. Zhitnitsky, Contact Term, its Holographic Description in QCD and Dark Energy, *Phys. Rev. D* 86 (2012) 045026. [arXiv:1112.3365](#), [doi:10.1103/PhysRevD.86.045026](#).
- [17] E. Thomas, A. R. Zhitnitsky, Topological Susceptibility and Contact Term in QCD. A Toy Model, *Phys. Rev. D* 85 (2012) 044039. [arXiv:1109.2608](#), [doi:10.1103/PhysRevD.85.044039](#).
- [18] F. R. Klinkhamer, G. E. Volovik, Vacuum energy density kicked by the electroweak crossover, *Phys. Rev. D* 80 (2009) 083001. [arXiv:0905.1919](#), [doi:10.1103/PhysRevD.80.083001](#).
- [19] G. E. Volovik, On spectrum of vacuum energy, *J. Phys. Conf. Ser.* 174 (2009) 012007. [arXiv:0801.2714](#), [doi:10.1088/1742-6596/174/1/012007](#).
- [20] K. K. Boddy, J. L. Feng, M. Kaplinghat, T. M. P. Tait, Self-Interacting Dark Matter from a Non-Abelian Hidden Sector, *Phys. Rev. D* 89 (11) (2014) 115017. [arXiv:1402.3629](#), [doi:10.1103/PhysRevD.89.115017](#).
- [21] A. Soni, Y. Zhang, Hidden SU(N) Glueball Dark Matter, *Phys. Rev. D* 93(11)(2016) 115025. [arXiv:1602.00714](#), [doi:10.1103/PhysRevD.93.115025](#).
- [22] J. Halverson, B. D. Nelson, F. Ruehle, String Theory and the Dark Glueball Problem, *Phys. Rev. D* 95 (4) (2017) 043527. [arXiv:1609.02151](#), [doi:10.1103/PhysRevD.95.043527](#).
- [23] J. Halverson, B. D. Nelson, F. Ruehle, G. Salinas, Dark Glueballs and their Ultralight Axions, *Phys. Rev. D* 98 (4) (2018) 043502. [arXiv:1805.06011](#), [doi:10.1103/PhysRevD.98.043502](#).
- [24] L. Forestell, D. E. Morrissey, K. Sigurdson, Cosmological Bounds on Non-Abelian Dark Forces, *Phys. Rev. D* 97 (7) (2018) 075029. [arXiv:1710.06447](#), [doi:10.1103/PhysRevD.97.075029](#).

- [25] P. Carenza, T. Ferreira, R. Pasechnik, Z.-W. Wang, Glueball dark matter, precisely, *Phys. Rev. D* 108 (12) (2023) 123027. [arXiv:2306.09510](#), [doi:10.1103/PhysRevD.108.123027](#).
- [26] E. D. Carlson, M. E. Machacek, L. J. Hall, Self-interacting dark matter, *Astrophys. J.* 398 (1992) 43–52. [doi:10.1086/171833](#).
- [27] Y. Hochberg, E. Kuflik, T. Volansky, J. G. Wacker, Mechanism for Thermal Relic Dark Matter of Strongly Interacting Massive Particles, *Phys. Rev. Lett.* 113 (2014) 171301. [arXiv:1402.5143](#), [doi:10.1103/PhysRevLett.113.171301](#).
- [28] Y. Hochberg, E. Kuflik, H. Murayama, T. Volansky, J. G. Wacker, Model for Thermal Relic Dark Matter of Strongly Interacting Massive Particles, *Phys. Rev. Lett.* 115 (2) (2015) 021301. [arXiv:1411.3727](#), [doi:10.1103/PhysRevLett.115.021301](#).
- [29] Y. Hochberg, E. Kuflik, H. Murayama, SIMP Spectroscopy, *JHEP* 05 (2016) 090. [arXiv:1512.07917](#), [doi:10.1007/JHEP05\(2016\)090](#).
- [30] D. Pappadopulo, J. T. Ruderman, G. Trevisan, Dark matter freeze-out in a nonrelativistic sector, *Phys. Rev. D* 94 (3) (2016) 035005. [arXiv:1602.04219](#), [doi:10.1103/PhysRevD.94.035005](#).
- [31] M. Farina, D. Pappadopulo, J. T. Ruderman, G. Trevisan, Phases of Cannibal Dark Matter, *JHEP* 12 (2016) 039. [arXiv:1607.03108](#), [doi:10.1007/JHEP12\(2016\)039](#).
- [32] E. Kuflik, M. Perelstein, N. R.-L. Lorier, Y.-D. Tsai, Elastically Decoupling Dark Matter, *Phys. Rev. Lett.* 116 (22) (2016) 221302. [arXiv:1512.04545](#), [doi:10.1103/PhysRevLett.116.221302](#).
- [33] S. W. Randall, M. Markevitch, D. Clowe, A. H. Gonzalez, M. Bradac, Constraints on the Self-Interaction Cross-Section of Dark Matter from Numerical Simulations of the Merging Galaxy Cluster 1E 0657-56, *Astrophys. J.* 679 (2008) 1173–1180. [arXiv:0704.0261](#), [doi:10.1086/587859](#).
- [34] M. Kaplinghat, S. Tulin, H.-B. Yu, Dark Matter Halos as Particle Colliders: Unified Solution to Small-Scale Structure Puzzles from Dwarfs to Clusters, *Phys. Rev. Lett.* 116 (4) (2016) 041302. [arXiv:1508.03339](#), [doi:10.1103/PhysRevLett.116.041302](#).

- [35] S. Tulin, H.-B. Yu, Dark Matter Self-interactions and Small Scale Structure, *Phys. Rept.* 730 (2018) 1–57. [arXiv:1705.02358](#), [doi:10.1016/j.physrep.2017.11.004](#).
- [36] T. Bernal, S. Capozziello, G. Cristofano, M. De Laurentis, MOND's acceleration scale as a fundamental quantity, *Mod. Phys. Lett. A* 26 (2011) 2677–2687. [arXiv:1110.2580](#), [doi:10.1142/S0217732311037042](#).